ATTICUS HOMESCHOOL

GRADE 6 | DAY 6

TUESDAY, SEPTEMBER 1, 2026

START TIME: 9:00 AM

TODAY'S GOAL

Finish Monday's required exit items, then strengthen ratios, precise reading/writing, and academic vocabulary while moving forward in science and French.

Today includes real breaks. Breaks are part of the schedule, not unfinished class time.

Lesson order is different from Monday.

Time	Lesson
9:00-9:15	Monday Closeout - finish 3 exit items
9:15-10:00	Writing - logical connectors + sentence boundaries
10:00-10:15	BREAK - snack, stretch, water
10:15-10:55	French - etre vs avoir + negation
11:00-12:00	Mathematics - equivalent ratios + ratio tables
12:00-12:45	LUNCH
12:45-1:30	History/Geography - culturally connected vs politically unified
1:30-2:10	Science - reflection, absorption, transmission + color
2:10-2:25	BREAK - outside / movement
2:25-3:10	Literature - motivation, theme, evidence
3:15-3:50	Computer Science / AI - training vs testing + generalization
3:50-4:00	BREAK - reset
4:00-4:25	Art / Design - complete visual explainer + closeout

9:00-9:15

MONDAY CLOSEOUT

Finish required exit items. No new teaching here.

WHY THIS COMES FIRST

These were required Monday exit items left blank. Completing them is not a punishment and is not a new assignment.

Use Monday's lesson pages if you need to reread the question. Do not use answer keys.

1. LITERATURE EXIT: What is the difference between a topic such as 'mistakes' and a theme about mistakes? Answer in 1-2 sentences.

2. HISTORY EXIT: In 3-4 sentences, compare Athens and Sparta. Include one similarity, one difference, and explain how the comparison shows Greek poleis could be culturally connected but politically independent.

3. CS/AI EXIT A: Explain why high model confidence does not guarantee a correct prediction.

4. CS/AI EXIT B: Name one change to training data that could improve generalization, and explain why it could help.

9:15-10:00

WRITING 6 - CONNECT IDEAS LOGICALLY

Because, but, so, and sentence boundaries.

WORDS YOU NEED TODAY

conjunction - a word that connects ideas, such as because, but, or so

cause - why something happens

effect - what happens as a result

contrast - a difference between ideas

run-on sentence - two or more complete thoughts joined without correct punctuation

comma splice - two complete sentences joined only with a comma

LISTEN FIRST - VOICE TUTOR PROMPT

Teach Atticus to choose a connector based on the relationship between ideas. BECAUSE introduces a reason, SO introduces a result, and BUT introduces a contrast. Connect this to Monday's sentence 'The classifier showed 99% confidence, so the prediction was incorrect' - explain that the grammar is possible but the logic is wrong because high confidence did not cause the error. Model three examples before practice. When correcting, name the exact relationship: cause, result, or contrast. Keep sentence mechanics separate from idea quality.

WORKED EXAMPLE 1 - CAUSE

Engineers add expansion joints BECAUSE metal changes size with temperature.

WORKED EXAMPLE 2 - RESULT

The key was missing, SO the team used the spare.

WORKED EXAMPLE 3 - CONTRAST

The classifier was 99% confident, BUT the prediction was wrong.

Guided practice

1. Choose BECAUSE, BUT, or SO: The wax paper transmits light, ______ the image is blurry because light is scattered.

2. Choose BECAUSE, BUT, or SO: The mountains made land travel difficult, ______ many Greek communities developed separately.

3. Repair this run-on using correct punctuation: 'Malik admitted the mistake the team found the spare key the test could continue.'

4. Explain why BUT is better than SO in this sentence: 'The model was 99% confident, ___ the prediction was wrong.'

9:15-10:00	WRITING 6 - INDEPENDENT Write clearly enough that the logic is visible.

1. Combine with BECAUSE: 'Transparent materials form clear images. They scatter very little transmitted light.'

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2. Combine with SO: 'The ratio 2:5 was doubled. The equivalent ratio became 4:10.'

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3. Combine with BUT: 'Athens and Sparta shared Greek culture. Their governments were different.'

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4. Repair the punctuation and connector: 'Priya was frustrated, so she still helped Malik find the spare key.' Choose the connector that best matches the meaning and rewrite the sentence.

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Short writing

Write 7-9 sentences explaining why a reader must understand the RELATIONSHIP between ideas, not just the facts themselves. Use at least two examples from school subjects today. Include one cause/effect example and one contrast example.

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10:15-10:55

FRENCH 6 - ETRE VS AVOIR

Choose the verb by meaning, then conjugate it.

WORDS YOU NEED TODAY

etre - to be

avoir - to have

meaning cue - the idea in the sentence that tells which verb belongs

conjugate - change the verb form to match the subject

negation - ne...pas or n'...pas

de after negation - in many 'have' sentences, un/une/des becomes de after ne...pas

LISTEN FIRST - VOICE TUTOR PROMPT

Teach the meaning difference before conjugation: ETRE describes being/state/identity; AVOIR expresses having/possessing and appears in common expressions. Repair Monday's error 'Ils sont deux livres' by asking whether the sentence means 'they ARE two books' or 'they HAVE two books.' Then model Nous sommes prets, Ils ont deux livres, Je n'ai pas de crayon. Keep French natural and continuing-level. Ask Atticus to say each sentence before writing it.

WORKED EXAMPLE 1

Nous sommes prets. = We are ready. State -> etre.

WORKED EXAMPLE 2

Ils ont deux livres. = They have two books. Possession -> avoir.

WORKED EXAMPLE 3

J'ai un crayon. -> Je n'ai pas de crayon.

Practice

1. Choose etre or avoir, then conjugate: Tu ______ fatigue aujourd'hui.

2. Choose etre or avoir, then conjugate: Nous ______ trois cahiers.

3. Choose etre or avoir, then conjugate: Elles ______ contentes.

4. Choose etre or avoir, then conjugate: Il ______ un chien.

5. Make negative: J'ai un stylo.

6. Make negative: Nous sommes en retard.

7. Answer in a complete French sentence: Est-ce que tu as des devoirs aujourd'hui?

8. Answer in a complete French sentence: Est-ce que tu es pret pour les maths?

ORAL CHECK

Say six short sentences: three with etre, three with avoir, including at least two negative sentences.

11:00-12:00

MATHEMATICS 6 - EQUIVALENT RATIOS

Keep the same relationship across every pair.

WORDS YOU NEED TODAY

ratio - a comparison of two quantities in a stated order

equivalent ratios - ratios that describe the same comparison

scale factor - the multiplier or divisor applied to BOTH quantities

ratio table - a table of equivalent ratio pairs

double number line - two aligned number lines used to show matching equivalent quantities

simplest form - a ratio divided by the greatest common factor so both numbers are as small as possible

LISTEN FIRST - VOICE TUTOR PROMPT

Monday showed that Atticus understands what a ratio is but does not yet preserve the same scale factor reliably in ratio tables. Do NOT introduce unit rate yet. Teach one invariant rule: whatever multiplier or divisor is applied to the first quantity must also be applied to the second quantity. Use 2 notebooks : 5 pencils and build 2:5, 4:10, 6:15, 8:20. Then revisit 5 cereal : 2 nuts so 15 cereal means multiply both by 3 -> 6 nuts. Use grid paper for tables and aligned work. Work three complete examples before independent practice.

WORKED EXAMPLE 1

2:5 -> multiply both by 2 -> 4:10. Same scale factor on both numbers.

WORKED EXAMPLE 2

2:5 -> multiply both by 3 -> 6:15. NOT 6:20, because 5 was multiplied by 3, not 4.

WORKED EXAMPLE 3

5 cereal : 2 nuts. For 15 cereal, multiply 5 by 3, so multiply 2 by 3 -> 6 nuts.

Guided practice - use the grid page

1. Complete the equivalent ratio table for 3 red beads : 4 blue beads: 3:4, 6:__, 9:__, 12:__. Write the scale factor for each row.

2. Complete: 5 notebooks : 2 folders. If there are 15 notebooks, how many folders? Show the scale factor.

3. Are 4:6 and 10:15 equivalent? Simplify BOTH ratios and compare.

4. A recipe uses 2 cups rice for 3 cups water. Complete a table for 2:3, 4:__, 6:__, 8:__. Explain the pattern in one sentence.

WORKSPACE

GRID PAPER - RATIOS / TABLES / DOUBLE NUMBER LINES

Use for ratio tables, double number lines, and organized scratch work.

Name: ____________________________ Date: September 1, 2026

11:00-12:00

MATHEMATICS 6 - INDEPENDENT + EXIT

Use one consistent scale factor.

1. Complete the table for 4 blue tiles : 7 white tiles: 4:7, 8:__, 12:__, 20:__. Show the scale factor for each pair.

2. A mixture uses 3 scoops powder for every 8 cups water. If 24 cups water are used, how many scoops powder are needed?

3. A map key uses 2 cm for every 15 km. Complete: 4 cm -> ___ km; 6 cm -> ___ km; 10 cm -> ___ km.

4. Are 6:9 and 14:21 equivalent? Simplify both ratios and explain.

5. A class has 12 students wearing sneakers and 8 wearing other shoes. Write sneakers:other in simplest form, then sneakers:total in simplest form.

EXIT TICKET - NO HELP

E1. Complete 5:8 = 15:____ and state the scale factor.

E2. A drink mix uses 2 parts concentrate for 7 parts water. If 21 parts water are used, how many parts concentrate are needed?

E3. Explain why 3:5 and 9:20 are not equivalent.

12:45-1:30

HISTORY & GEOGRAPHY - CULTURE VS POLITICAL UNITY

Answer Monday's question directly.

WORDS YOU NEED TODAY

culturally connected - sharing language, religion, customs, stories, festivals, or other cultural practices

politically unified - governed together under one political system or central authority

polis - an independent Greek city-state and its surrounding territory

independent - governing itself

shared identity - a sense of belonging created by common culture or traditions

rivalry - competition or conflict between groups

LISTEN FIRST - VOICE TUTOR PROMPT

Start by answering Atticus's Monday closeout question directly: culturally connected means groups share culture; politically unified means they are governed together by one political system. Use a modern analogy that does not distort history: different independent countries may share a language or cultural tradition without sharing one government. Then return to Athens and Sparta. Explicitly show: shared Greek language/religion/festivals = cultural connection; separate governments and laws = political independence. Ask self-contained comparison questions.

WORKED EXAMPLE

Athens and Sparta both participated in Greek religious traditions, but each polis had its own government. That means culturally connected, not politically unified.

Mini-reading - Shared Greek culture, separate governments

Greek poleis often shared language, religious stories, sanctuaries, festivals, and athletic contests. Travelers could recognize many customs across the Greek world. Yet each polis could make its own laws, organize its own government, train its own soldiers, and compete with neighboring poleis. Cultural connection therefore did not erase political independence. Athens and Sparta are a useful example: both were Greek, but they developed different political institutions and social priorities.

12:45-1:30

HISTORY & GEOGRAPHY - QUESTIONS

Use the exact distinction from the reading.

1. Give TWO examples of cultural connections that Greek poleis could share.

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2. Give TWO examples of political decisions a polis could control for itself.

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3. Explain why sharing the same gods or festivals does NOT prove that Athens and Sparta had one government.

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4. Complete the sentence accurately: Athens and Sparta were culturally connected because ________, but politically independent because ________.

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5. Why can culturally connected societies still become rivals? Use one plausible reason from the Greek polis context.

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EXIT TICKET

In 3 sentences: (1) define culturally connected, (2) define politically unified, and (3) explain which description best fits Athens and Sparta and why.

1:30-2:10

SCIENCE 6 - REFLECTION, ABSORPTION, TRANSMISSION

Move forward while keeping the clarity test.

WORDS YOU NEED TODAY

reflect - bounce light away from a surface

absorb - take in light energy

transmit - let light pass through

scatter - redirect light in many directions

wavelength - a property of light related to color

visible light - the part of light humans can see

LISTEN FIRST - VOICE TUTOR PROMPT

Monday showed that material classification is now much stronger. Begin with a 3-minute check: transparent versus translucent depends on image clarity/scattering, not just whether light passes. Then move forward: when light hits matter, some may be reflected, absorbed, or transmitted. Introduce color at a simple Grade 6 level: an object appears a color because the light reaching the eye contains wavelengths reflected from the object; other visible wavelengths may be absorbed. Avoid saying an object 'contains' its color. Use familiar examples like a red folder under white light.

WORKED EXAMPLE 1

Clear glass: much visible light is transmitted, so you can often see through it clearly.

WORKED EXAMPLE 2

Black paper: much visible light is absorbed, so little is reflected to your eye.

WORKED EXAMPLE 3

A red object under white light reflects more red light toward your eyes and absorbs much of the other visible light.

Guided practice

1. A mirror looks bright because it reflects much of the light that hits it. Which interaction is most important here: reflection, absorption, or transmission?

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2. A black shirt warms in sunlight partly because it absorbs light energy. Which interaction is described?

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3. A clear window lets you see outside. Which interaction allows light from outside to reach your eyes through the glass?

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4. Why does saying 'wax paper transmits light' not automatically mean wax paper is transparent?

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5. Under white light, why can a red book look red to your eyes?

EXIT TICKET

E1. Name the three main light-matter interactions in today's lesson.

E2. In one sentence, explain why a red object can appear red under white light.

2:25-3:10	LITERATURE - MOTIVATION, THEME, AND EVIDENCE
	Evidence must prove the idea, not merely mention the plot.

WORDS YOU NEED TODAY

motivation - the reason a character wants or chooses something

inference - a conclusion based on evidence plus reasoning

theme - a complete message about life, choices, or human experience

evidence - a specific detail that supports an interpretation

relevant evidence - evidence that directly helps prove the idea

objective summary - a brief account of main events without opinion

LISTEN FIRST - VOICE TUTOR PROMPT

Monday showed good literal comprehension but weaker motivation and theme evidence. Explicitly repair 'he has no motivation whatsoever': Malik was motivated to avoid embarrassment, blame, or delay. Teach that motivation answers WHY a character considers an action. Then teach evidence quality: 'Malik lost the key' alone does not strongly prove a theme about responsibility; his admission, the team's spare-key system, and the checklist changes are stronger evidence. Model one unrelated example before today's story.

Original story - The Unsent Message

Nora had volunteered to send the robotics club's registration form before the Friday deadline. She filled in the team names, attached the design summary, and saved the draft email. Then her phone rang, and she left the computer to help her younger brother with dinner.

On Monday morning, her teammate Luis asked whether the organizers had replied. Nora opened the sent folder. The message was not there.

For a moment, she considered saying the website must have failed. If everyone believed it was a technical problem, she would not have to admit that she had never pressed Send.

Instead, Nora showed Luis the draft. 'I saved it, but I didn't send it. I messed up.'

Luis looked worried, but he did not argue. He opened the competition page and found a contact address for late-registration questions. Together they wrote a short explanation and attached the time-stamped draft. The organizers allowed the club to register because the form had been completed before the deadline.

That afternoon, the club changed its process. Important submissions would now require two people to confirm that the message appeared in the sent folder. Nora added the first checkmark beside her own name.

'You don't trust me anymore?' she asked.

'I trust you,' Luis said. 'The checklist is for all of us.'

2:25-3:10

LITERATURE - CLOSE READING

Answer WHY, then support it with the best detail.

1. What mistake did Nora make before Monday morning?

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2. Nora considers blaming a technical failure. What is her likely motivation for considering that explanation? Use one detail from the story.

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3. What does Nora ultimately do instead? What does that choice reveal about her character?

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4. Why is the time-stamped draft important to the plot?

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5. Write a universal theme statement about responsibility, trust, or systems. Do not use a character's name.

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6. Give TWO pieces of evidence that directly support your theme. For each one, add one sentence explaining what it proves.

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7. Which is stronger evidence for a theme about responsibility: 'Nora forgot to press Send' OR 'Nora admitted the mistake and helped create a two-person confirmation system'? Explain why.

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8. Write a 3-sentence objective summary.

EXIT TICKET

Explain the difference between a character's motivation and a story's theme.

3:15-3:50

COMPUTER SCIENCE / AI - TRAINING VS TESTING

Good models must work beyond the exact training examples.

WORDS YOU NEED TODAY

training data - examples used to fit or teach a model

test data - examples used to evaluate how the model performs after training

generalization - working well on new examples or conditions

confidence - how strongly the model matches an input to one of its learned classes

correctness - whether the prediction is actually right

overfitting - performing well on familiar training patterns but poorly on new ones

LISTEN FIRST - VOICE TUTOR PROMPT

Monday showed that Atticus still needs the difference between varied training data and repeated nearly identical examples. Teach training versus testing and a simple idea of overfitting. Use an example: 30 cup photos from the same angle/background may let the model memorize a narrow pattern; varied angles/backgrounds are more likely to help generalization. Re-emphasize that confidence must be interpreted together with correctness. Ask questions that always state the actual object, the prediction, and the confidence when relevant.

WORKED EXAMPLE 1

Training set: 30 nearly identical cup photos. Risk: the model may learn the white table or one viewing angle rather than the broader idea of a cup.

WORKED EXAMPLE 2

More varied training: cups at different angles, distances, and backgrounds. This usually gives the model a better chance to generalize.

3:15-3:50

COMPUTER SCIENCE / AI - PRACTICE

Training data teaches; test data evaluates.

1. Which is training data: the examples used to teach the model, or the new examples used afterward to evaluate it?

2. Why can 30 nearly identical cup photos be less useful for generalization than 30 varied cup photos?

3. A model predicts CUP for an actual BOOK with 99% confidence. State (a) the prediction, (b) the actual object, (c) whether the prediction is correct, and (d) what 99% confidence does and does not mean.

4. A classifier is perfect on its training images but makes many mistakes on new backgrounds. What problem might this suggest?

EXIT TICKET

E1. Define generalization in your own words.

E2. Explain one difference between training data and test data.

E3. Name one way to reduce the risk that a simple image classifier learns only the background.

4:00-4:25	ART / DESIGN - FINISH THE VISUAL EXPLAINER
	Turn Monday's plan into a real portfolio piece.

WORDS YOU NEED TODAY

visual hierarchy - the order in which a viewer notices information

annotation - a label or note that explains part of a visual

contrast - a difference that helps important information stand out

composition - the arrangement of visual elements

caption - brief text that explains the visual

LISTEN FIRST - VOICE TUTOR PROMPT

Atticus already planned a visual explainer on transparent, translucent, and opaque materials. Today he must create the FINAL piece, not another plan. Use his plan: flashlight shining through three materials; labels explaining clear image / blurry image / no transmitted visible light. Ask him what the viewer should notice first, then let him design it. Require scientific accuracy: translucent transmits some light but scatters it; opaque does not transmit visible light. Do not design it for him.

FINAL PIECE REQUIREMENTS

Title: Transparent, Translucent, Opaque

One flashlight/light-source diagram across three materials

At least 3 labels/annotations

Show clear image vs blurry image vs blocked light

Use size/contrast so the three categories are easy to compare

2-3 sentence caption using the words transmit and scatter

Name + date

Use the second grid sheet OR a blank sheet if you want a cleaner final design.

WORKSPACE

GRID / DESIGN PLANNING PAGE

Use for ratio tables, double number lines, and organized scratch work.

Name: ______________________ Date: September 1, 2026

4:20-4:25

TUESDAY CLOSEOUT

Started -> Finished -> Checked -> Filed

NOTHING MISSING CHECK

Monday closeout exits

Writing independent + short writing

French practice + oral check

Math guided + independent + exit

History questions + exit

Science guided + exit

Literature questions + exit

CS/AI questions + exit

Art/Design FINAL piece

One thing I understand better today:

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One glossary word I can define without looking:

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One question I still have:

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PARENT

TEACHER KEY + TUESDAY PACING NOTES

Keep separate from student pages.

Monday evidence used

Monday showed clear improvement in science material classification; ratio meaning is emerging but ratio tables/equivalent scaling are inconsistent; French needs etre/avoir meaning repair; History needs an explicit cultural-connection vs political-unity distinction; Literature needs stronger motivation and theme evidence; Writing needs logic-aware conjunctions and sentence boundaries; CS/AI needs training/testing/generalization reinforcement. Several Monday exit items were blank, so Tuesday begins with a short completion block rather than assigning zeros.

Writing: 1 BUT. 2 SO. 3 Accept sentences with correct boundaries, e.g., Malik admitted the mistake. The team found the spare key, so the test could continue. 4 BUT marks contrast; SO would incorrectly imply confidence caused the error.

French: 1 es. 2 avons. 3 sont. 4 a. 5 Je n'ai pas de stylo. 6 Nous ne sommes pas en retard. Open responses vary.

Math guided: 1 6:8, 9:12, 12:16 with factors 2,3,4. 2 6 folders. 3 yes: 4:6=2:3 and 10:15=2:3. 4 4:6, 6:9, 8:12.

Math independent: 1 8:14, 12:21, 20:35. 2 9 scoops. 3 30 km, 45 km, 75 km. 4 yes, both 2:3. 5 sneakers:other=3:2; sneakers:total=3:5.

Math exit: E1 24, factor 3. E2 6 parts concentrate. E3 3:5=0.6 while 9:20=0.45; no common scale factor.

History: Cultural connection examples: shared language/religion/festivals/stories. Political independence: own laws/government/military. Athens and Sparta were culturally connected but not politically unified.

Science: 1 reflection. 2 absorption. 3 transmission. 4 because translucent can transmit while scattering; transparency also requires a clear image. 5 red wavelengths are reflected toward the eyes more strongly than many others. Exit: reflect/absorb/transmit; red appearance from reflected red light reaching eyes.

Literature: Motivation: avoid embarrassment/blame/delay. Strong theme evidence includes Nora admitting the mistake, using the time-stamped draft, and helping create a two-person confirmation system. Stronger evidence is the admission/system change because it directly shows responsibility and learning.

CS/AI: 1 training data = teaching examples. 2 varied data covers more conditions. 3 prediction CUP, actual BOOK, incorrect, 99% = strong model preference among known classes not a guarantee. 4 overfitting / narrow training. Exit: generalization = works on new examples; training fits model, test evaluates it; vary backgrounds/angles or include diverse examples.

PARENT

PACING AFTER TUESDAY

Use Tuesday evidence to decide Wednesday.

Pacing after Tuesday

Do not move to unit rate unless ratio exit is secure and ratio-table errors disappear. Science may continue deeper into light/color if exit is accurate. French can add aller only after etre/avoir choice is reliable. History can progress once culturally connected vs politically unified is stated accurately without prompting. Literature should stay on motivation/theme/evidence until he reliably selects evidence that proves the theme. CS/AI should not add new ML vocabulary if training/test/generalization remains fuzzy.